

Topic D HL - Paper 1A - Study Guide [43 marks]

1. [Maximum mark: 1]

23M.1.SL.TZ1.23

The radius of the Earth is R . A satellite is launched to a height $h = \frac{R}{4}$ above the Earth's surface.

What is $\frac{\text{gravitational force on satellite at the surface}}{\text{gravitational force on satellite at height } h}$?

A. $\frac{4}{5}$

B. $\frac{16}{25}$

C. $\frac{25}{16}$

D. $\frac{5}{4}$

[1]

Markscheme

C

Examiners report

With a lower discrimination index, this question proved somewhat tricky for SL students. Option C was correctly chosen most frequently, however students were also distracted by options B and D.

2. [Maximum mark: 1]

23M.1.SL.TZ2.26

The gravitational field strength at the surface of the Earth is often taken to be 9.8 N kg^{-1} .

The use of this value to calculate the weight of an object **above** the surface of the Earth is

- A. a paradigm shift in our understanding of gravity.
- B. an attempt to model gravitational fields.
- C. an outcome from a peer review.
- D. an approximation used for estimation purposes.

[1]

Markscheme

D

3. [Maximum mark: 1]

23M.1.HL.TZ1.30

Two isolated point masses, P of mass m and Q of mass $2m$, are separated by a distance $3d$. X is a point a distance d from P and $2d$ from Q.



What is the net gravitational field strength at X and the net gravitational potential at X?

	Net gravitational field strength at X	Net gravitational potential at X
A.	$\frac{Gm}{d^2}$	0
B.	$\frac{Gm}{d^2}$	$-\frac{Gm}{d}$
C.	$\frac{Gm}{2d^2}$	0
D.	$\frac{Gm}{2d^2}$	$-\frac{2Gm}{d}$

Markscheme

D

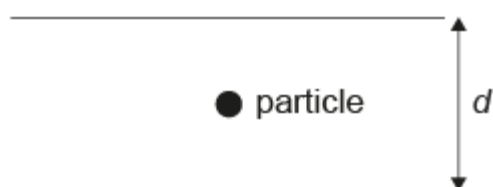
Examiners report

This question was particularly difficult for students. In fact, the correct answer (D) was the least common choice among the four options. Given the low discrimination index, it seems likely that students resorted to guessing with this question. As a result, this question would be a useful teaching tool when discussing how overlapping fields and potentials combine.

4. [Maximum mark: 1]

23M.1.HL.TZ1.31

A negatively charged particle is stationary halfway between two horizontal charged plates. The plates are separated by a distance d with potential difference V between them.



What is the magnitude of the electric field and direction of the electric field at the position of the particle?

	Magnitude of electric field	Direction of electric field
A.	$\frac{2V}{d}$	up

B.	$\frac{V}{d}$	up
C.	$\frac{2V}{d}$	down
D.	$\frac{V}{d}$	down

[1]

Markscheme

D

Examiners report

Option B was (incorrectly) chosen most frequently, with option D a close second. Students appeared comfortable determining the magnitude of the electric field in the example given, but not its direction.

5. [Maximum mark: 1]

23M.1.HL.TZ1.32

The escape speed from the surface of earth is v_{esc} . The radius of earth is R . A satellite of mass m is in orbit at a height $\frac{R}{4}$ above the surface of the Earth. What is the energy required to move the satellite to infinity?

A. $\frac{mv_{\text{esc}}^2}{5}$

B. $\frac{2mv_{\text{esc}}^2}{5}$

C. mv_{esc}^2

D. $2mv_{\text{esc}}^2$

[1]

Markscheme

A

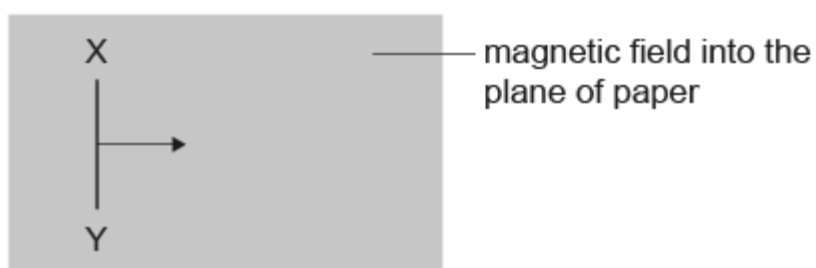
Examiners report

This question had the lowest difficulty index in the paper, suggesting that students found this question particularly tricky. However, with such a low discrimination index, once again students appear to have been guessing. The correct option (A) was the least frequent answer selected.

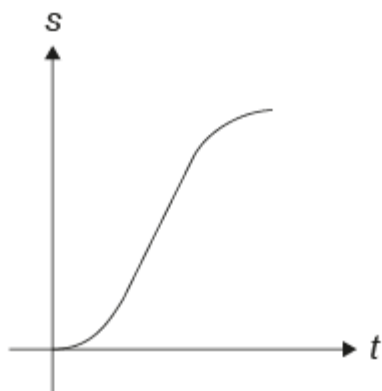
6. [Maximum mark: 1]

23M.1.HL.TZ1.34

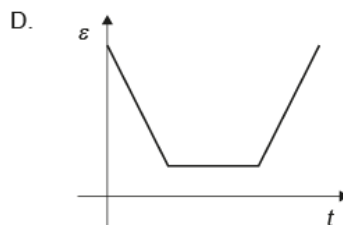
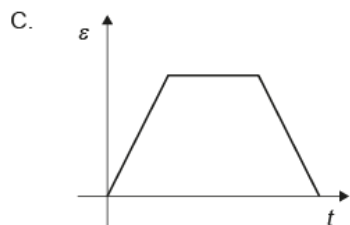
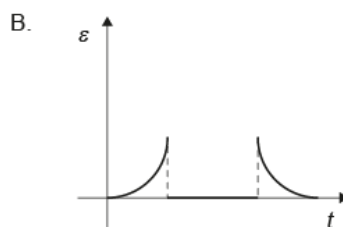
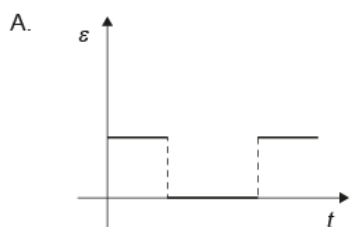
Wire XY moves perpendicular to a magnetic field in the direction shown.



The graph shows the variation with time of the displacement of XY.



What is the graph of the electromotive force (emf) ε induced across XY?



[1]

Markscheme

C

Examiners report

This question was well answered by HL students.

7. [Maximum mark: 1]

23M.1.HL.TZ2.20

The gravitational field strength at the surface of the Earth is often taken to be 9.8 N kg^{-1} .

The use of this value to calculate the weight of an object **above** the surface of the Earth is

- A. a paradigm shift in our understanding of gravity.
- B. an attempt to model gravitational fields.

C. an outcome from a peer review.

[1]

D. an approximation used for estimation purposes.

Markscheme
D

8. [Maximum mark: 1]

23M.1.HL.TZ2.31

The mass of Mars is about ten times that of the Moon. The radius of Mars is about twice that of the Moon.

What is the $\frac{\text{escape speed from Mars}}{\text{Moon}}$?

A. $\sqrt{5}$

B. $2\sqrt{5}$

C. 5

D. 25

[1]

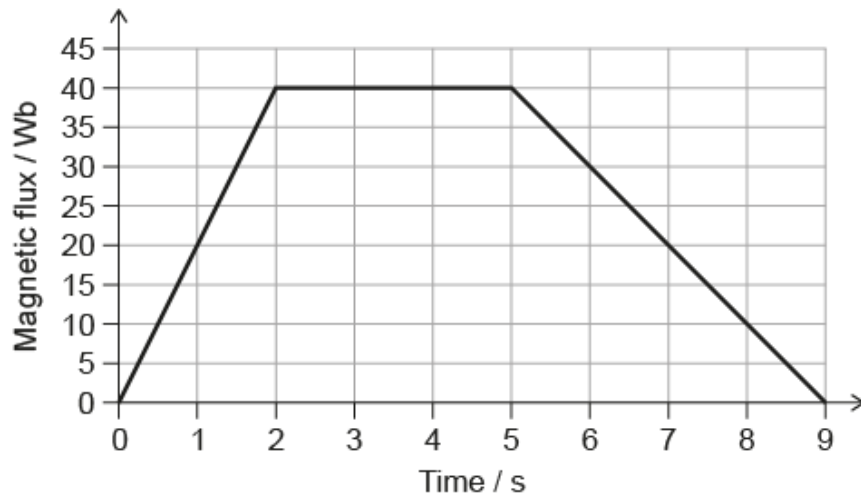
Markscheme
A

9. [Maximum mark: 1]

23M.1.HL.TZ2.32

A single loop of wire of resistance $10\ \Omega$ has its plane perpendicular to a changing magnetic field.

The graph shows the variation with time of the magnetic flux linked through the loop of wire.



What is the maximum current in the loop of wire?

- A. 1.0 A
- B. 2.0 A
- C. 4.0 A
- D. 20 A

[1]

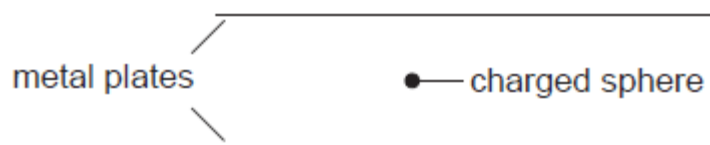
Markscheme

B

10. [Maximum mark: 1]

22M.1.HL.TZ1.32

A charged sphere in a gravitational field is initially stationary between two parallel metal plates. There is a potential difference V between the plates.



Three changes can be made:

- I. Increase the separation of the metal plates
- II. Increase V
- III. Apply a magnetic field into the plane of the paper

What changes made separately will cause the charged sphere to accelerate?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

[1]

Markscheme
A

Examiners report
Option C was a very successful distractor, selected by the majority of candidates. Most candidates missed that change III ("Apply a magnetic field into the plane of the paper") can never be correct if the charge is stationary.

11. [Maximum mark: 1]

22M.1.HL.TZ1.33

An object of mass m is launched from the surface of the Earth. The Earth has a mass M and radius r . The acceleration due to gravity at the surface of the Earth is g . What is the escape speed of the object from the surface of the Earth?

- A. $\sqrt{gr}$

B. $\sqrt{2gr}$

C. $\sqrt{2Mgr}$

D. $\sqrt{2mgr}$

[1]

Markscheme

B

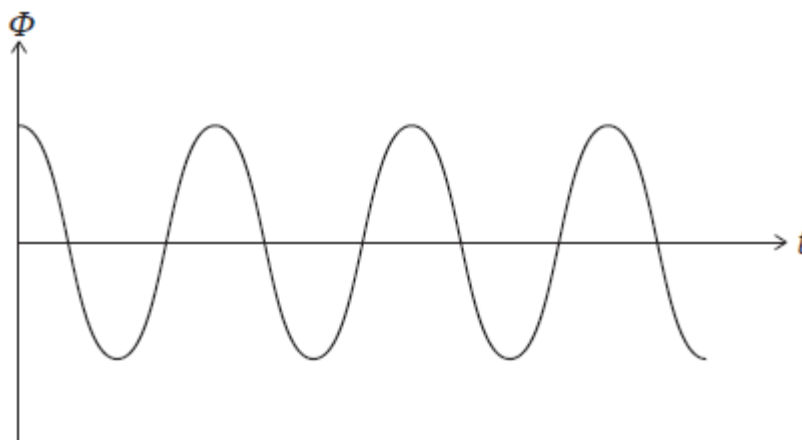
Examiners report

Options B and C were selected by a roughly equal number of candidates. Again, this is a situation where unit analysis is beneficial; options C and D would not produce units associated with speed (mass is already incorporated in the constant 'g').

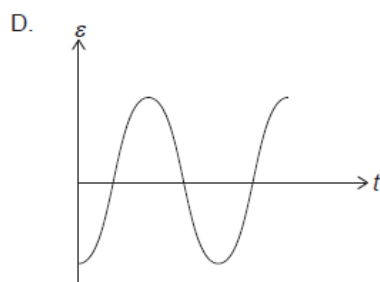
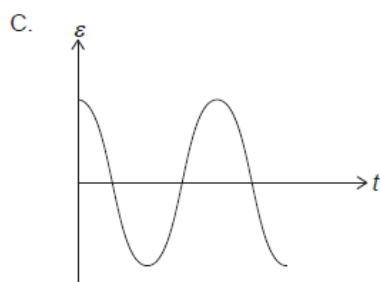
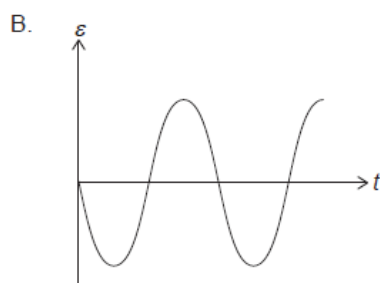
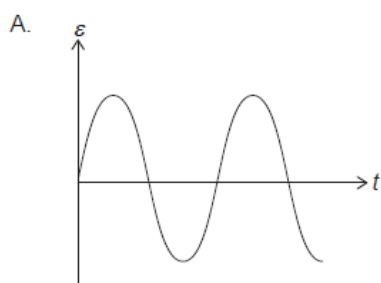
12. [Maximum mark: 1]

22M.1.HL.TZ1.34

The graph shows the variation of magnetic flux Φ in a coil with time t .



What represents the variation with time of the induced emf $\mathcal{E}$ across the coil?



[1]

Markscheme

A

Examiners report

Candidate selections were divided across the options presented, with option B as the most common (incorrect answer). This suggests that candidates could use more guidance on how to interpret slope on a ϵ vs. t graph.

13. [Maximum mark: 1]

22M.1.HL.TZ2.32

Two positive and two negative charges are located at the corners of a square as shown. Point X is the centre of the square. What is the value of the electric field E and the electric potential V at X due to the four charges?

$+Q$ ● $-Q$ ●

×

$-Q$ ● $+Q$ ●

	Electric field E at X	Electric potential V at X
A.	$E = 0$	$V = 0$
B.	$E \neq 0$	$V = 0$
C.	$E = 0$	$V \neq 0$
D.	$E \neq 0$	$V \neq 0$

[1]

Markscheme

A

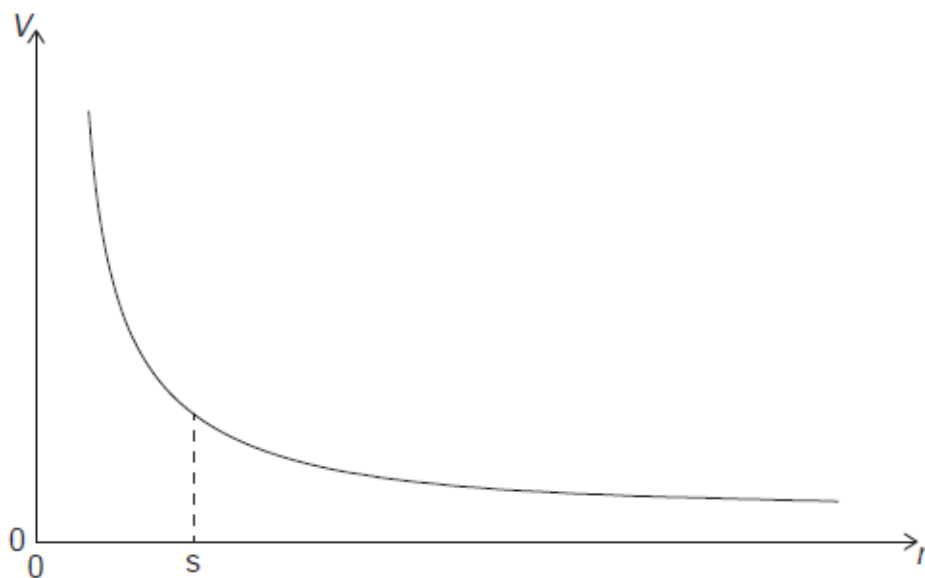
Examiners report

Candidates were unsure about this question with almost equal numbers choosing A and C. Electric potential is a scalar quantity so unaffected by the sign of the charge and can only be 0 in this arrangement removing the choice of C.

14. [Maximum mark: 1]

22M.1.HL.TZ2.33

The graph shows the variation with distance r of the electric potential V from a charge Q .



What is the electric field strength at distance s ?

- A. The area under the graph between s and infinity
- B. The area under the graph between 0 and s
- C. The gradient of the tangent at s
- D. The negative of the gradient of the tangent at s

[1]

Markscheme

D

15. [Maximum mark: 1]

22M.1.HL.TZ2.35

A conducting bar with vertices PQRS is moving vertically downwards with constant velocity v through a horizontal magnetic field B that is directed into the plane of the page.



Which side of the bar will have the greatest density of electrons?

- A. PQ
- B. QR
- C. RS
- D. SP

[1]

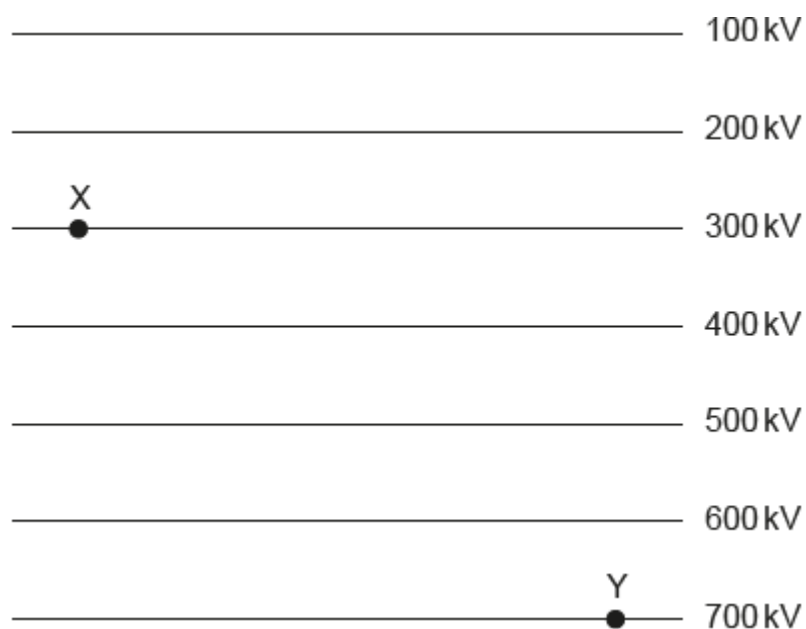
Markscheme

D

16. [Maximum mark: 1]

21M.1.HL.TZ1.30

A particle with charge $-2.5 \times 10^{-6} \text{ C}$ moves from point X to point Y due to a uniform electrostatic field. The diagram shows some equipotential lines of the field.



[1]

What is correct about the motion of the particle from X to Y and the magnitude of the work done by the field on the particle?

	Motion of the particle from X to Y	Magnitude of the work done by the field on the particle
A.	uniform linear	0 J
B.	uniform linear	1 J
C.	uniformly accelerated	0 J
D.	uniformly accelerated	1 J

Markscheme

D

17. [Maximum mark: 1]

21M.1.HL.TZ1.31

Which is a correct unit for gravitational potential?

A. m^2s^{-2}

B. J kg

C. m s^{-2}

D. $\text{N m}^{-1} \text{kg}^{-1}$

[1]

Markscheme
A

18. [Maximum mark: 1]

21M.1.HL.TZ1.32

A planet has radius R . The escape speed from the surface of the planet is v . At what distance from the surface of the planet is the orbital speed $0.5v$?

A. $0.5R$

B. R

C. $2R$

D. $4R$

[1]

Markscheme
B

19. [Maximum mark: 1]

21M.1.HL.TZ1.33

A conducting ring encloses an area of 2.0 cm^2 and is perpendicular to a magnetic field of strength 5.0 mT . The direction of the magnetic field is reversed in a time 4.0 s . What is the average emf induced in the ring?

A. 0

B. $0.25 \mu\text{V}$

C. $0.40 \mu\text{V}$

D. $0.50\ \mu\text{V}$

[1]

Markscheme
D

20. [Maximum mark: 1]

21M.1.HL.TZ1.34

The conservation of which quantity explains Lenz's law?

A. Charge

B. Energy

C. Magnetic field

D. Mass

[1]

Markscheme
B

21. [Maximum mark: 1]

21M.1.HL.TZ2.30

An object of mass m released from rest near the surface of a planet has an initial acceleration z . What is the gravitational field strength near the surface of the planet?

A. z

B. $\frac{z}{m}$

C. mz

D. $\frac{m}{z}$

[1]

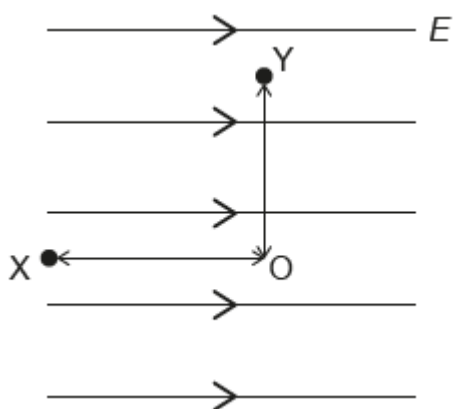
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Markscheme
A

22. [Maximum mark: 1]

21M.1.HL.TZ2.31

The points X and Y are in a uniform electric field of strength E . The distance OX is x and the distance OY is y .



What is the magnitude of the change in electric potential between X and Y?

A. Ex

B. Ey

C. $E(x + y)$

D. $E\sqrt{x^2 + y^2}$

[1]

Markscheme
A

23. [Maximum mark: 1]

21M.1.HL.TZ2.32

A satellite orbits planet **X** with a speed v_X at a distance r from the centre of planet **X**. Another satellite orbits planet **Y** at a speed of v_Y at a distance r from the centre of planet **Y**. The mass of planet **X** is M and the mass of planet **Y** is $4M$. What is the ratio of $\frac{v_X}{v_Y}$?

- A. 0.25
- B. 0.5
- C. 2.0
- D. 4.0

[1]

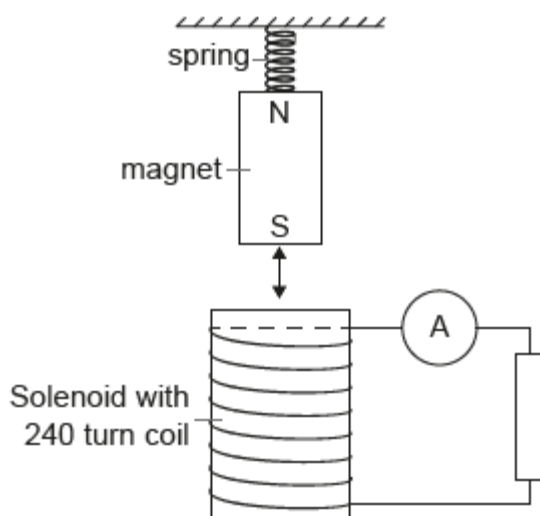
Markscheme

B

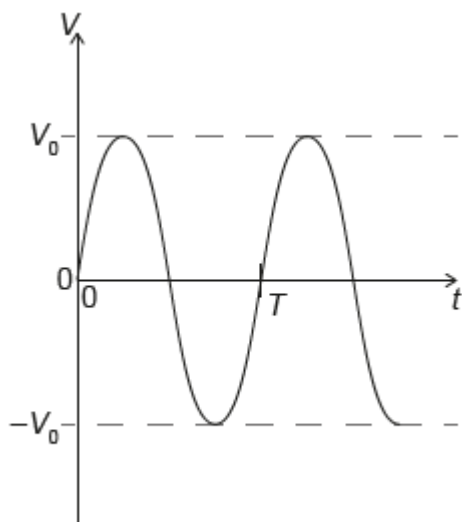
24. [Maximum mark: 1]

21M.1.HL.TZ2.35

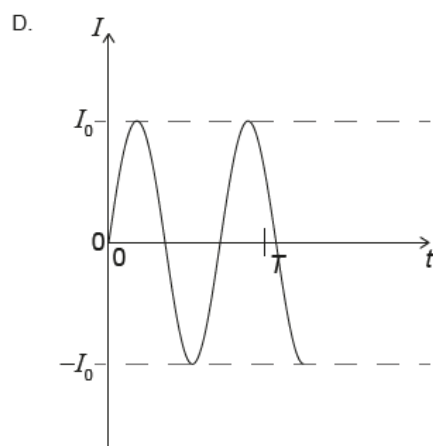
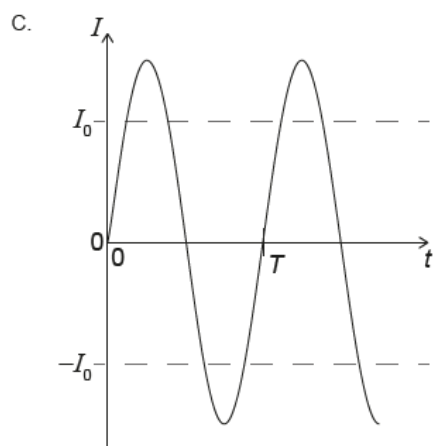
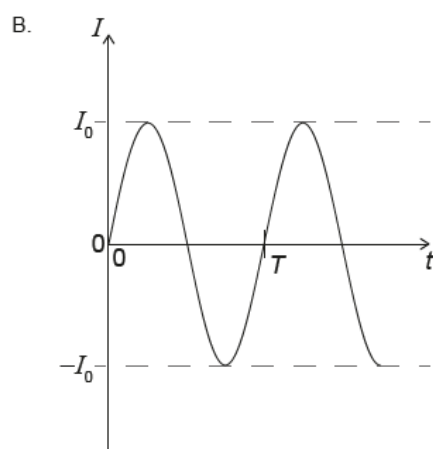
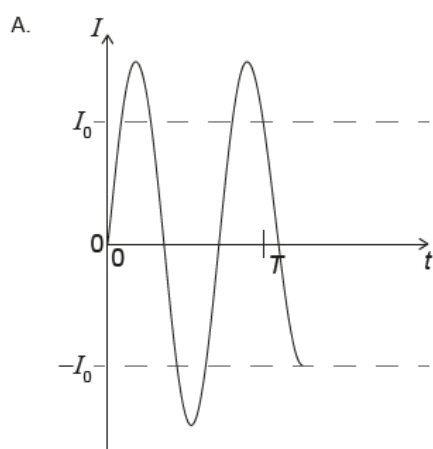
A magnet connected to a spring oscillates above a solenoid with a 240 turn coil as shown.



The graph below shows the variation with time t of the emf across the solenoid with the period, T , of the system shown.



The spring is replaced with one that allows the magnet to oscillate with a higher frequency. Which graph shows the new variation with time t of the current I in the resistor for this new set-up?



[1]

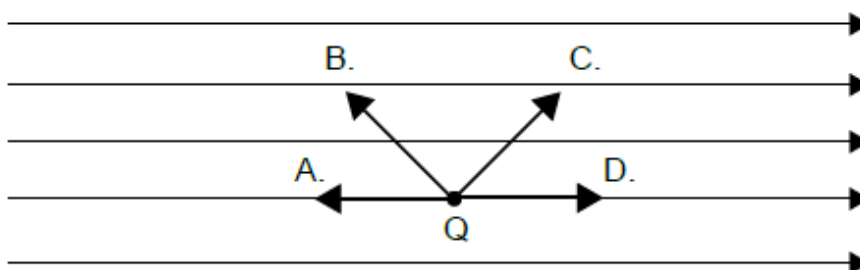
Markscheme
A

25. [Maximum mark: 1]

19M.1.HL.TZ1.32

A negative charge Q is to be moved within an electric field E , to equidistant points from its position, as shown.

Which path requires the most work done?



[1]

Markscheme
D

Examiners report
The most common answer was A, suggesting that students missed the prompt that Q is a negative charge.

26. [Maximum mark: 1]

19M.1.HL.TZ1.33

The escape velocity for an object at the surface of the Earth is v_{esc} . The diameter of the Moon is 4 times smaller than that of the Earth and the

mass of the Moon is 81 times smaller than that of the Earth. What is the escape velocity of the object on the Moon?

A. $\frac{2}{81} v_{\text{esc}}$

B. $\frac{4}{81} v_{\text{esc}}$

C. $\frac{2}{9} v_{\text{esc}}$

D. $\frac{4}{9} v_{\text{esc}}$

[1]

Markscheme

C

Examiners report

This question was well answered by candidates.

27. [Maximum mark: 1]

19M.1.HL.TZ1.34

A satellite in a circular orbit around the Earth needs to reduce its orbital radius.

What is the work done by the satellite rocket engine and the change in kinetic energy resulting from this shift in orbital height?

	Work done by the satellite rocket engine	Kinetic energy
A.	positive	increase
B.	positive	decrease
C.	negative	increase
D.	negative	decrease

[1]

Markscheme

C

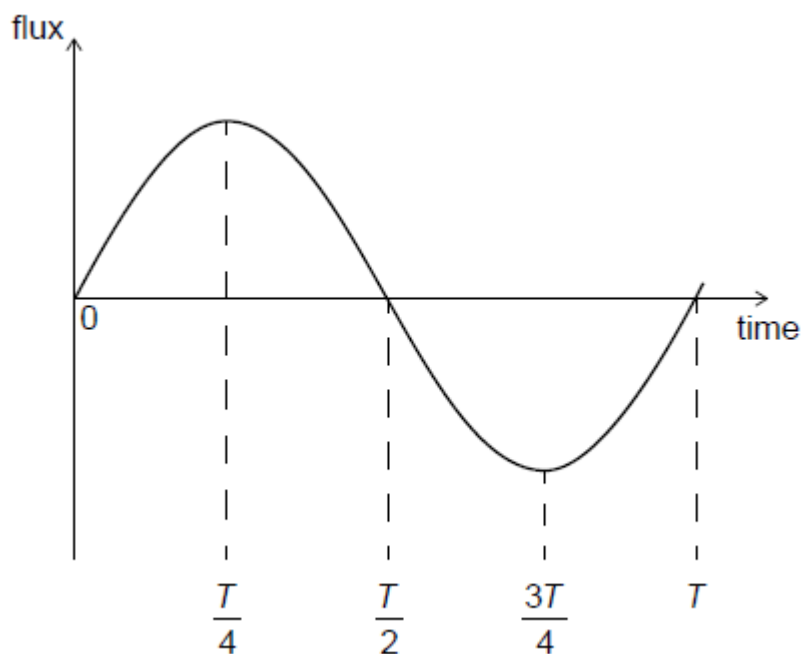
Examiners report

This question was generally well answered, however a significant number of students (incorrectly) selected response A suggesting a lack of clarity around the work done as a result of changes in orbital height.

28. [Maximum mark: 1]

19M.1.HL.TZ1.35

The graph below shows the variation with time of the magnetic flux through a coil.



Which of the following gives three times for which the magnitude of the induced emf is a maximum?

A. $0, \frac{T}{4}, \frac{T}{2}$

B. $0, \frac{T}{2}, T$

C. $0, \frac{T}{4}, T$

D. $\frac{T}{4}, \frac{T}{2}, \frac{3T}{4}$

[1]

Markscheme

B

Examiners report

This question was extremely well done, with the highest difficulty index seen on this paper.

29. [Maximum mark: 1]

19M.1.HL.TZ2.10

Satellite X is in orbit around the Earth. An identical satellite Y is in a higher orbit. What is correct for the total energy and the kinetic energy of the satellite Y compared with satellite X?

	Total energy of satellite X	Kinetic energy of satellite X
A.	larger	larger
B.	smaller	larger
C.	larger	smaller
D.	smaller	smaller

[1]

Markscheme

B

Examiners report

We accept the comment from G2 forms that the wording of this question could be improved. The correct answer (B) considers the total and kinetic energies of satellite X the most popular answer.

30. [Maximum mark: 1]

19M.1.HL.TZ2.11

The escape speed from a planet of radius R is v_{esc} . A satellite orbits the planet at a distance R from the surface of the planet. What is the orbital speed of the satellite?

A. $\frac{1}{2}v_{\text{esc}}$

B. $\frac{\sqrt{2}}{2}v_{\text{esc}}$

C. $\sqrt{2}v_{\text{esc}}$

D. $2v_{\text{esc}}$

[1]

Markscheme

A

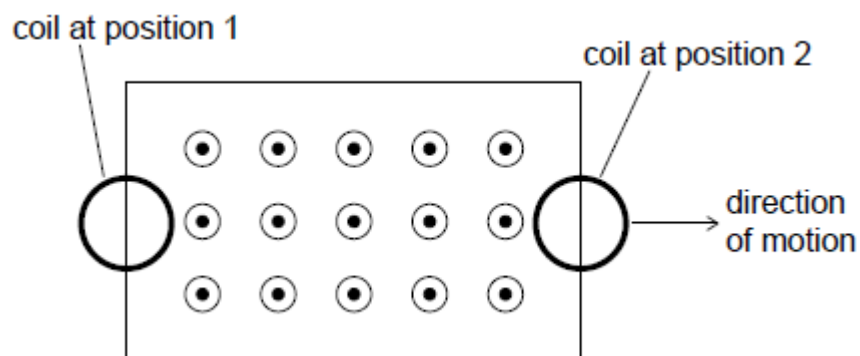
Examiners report

This had a very low discrimination index with the majority of candidates choosing B, followed by C. Response A, the correct answer, was third in popularity. The candidates missed that the satellite orbits at a distance of R from the surface of a planet of radius R so the total distance to be considered was $2R$.

31. [Maximum mark: 1]

19M.1.HL.TZ2.29

A circular coil of wire moves through a region of uniform magnetic field directed out of the page.



What is the direction of the induced conventional current in the coil for the marked positions?

	Position 1	Position 2
A.	clockwise	clockwise
B.	counterclockwise	clockwise
C.	clockwise	counterclockwise
D.	counterclockwise	counterclockwise

[1]

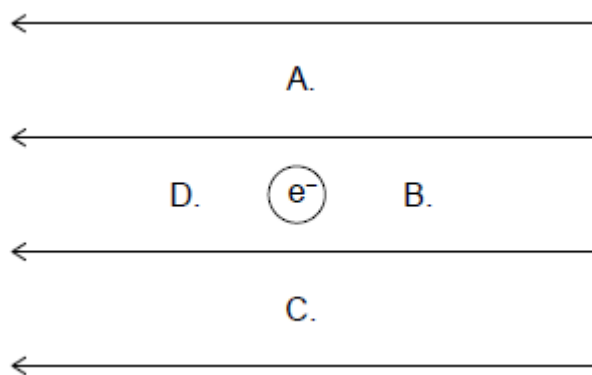
Markscheme

C

32. [Maximum mark: 1]

19M.1.HL.TZ2.30

An electron is fixed in position in a uniform electric field. What is the position for which the electrical potential energy of the electron is greatest?



[1]

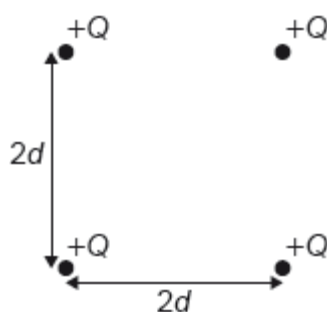
Markscheme

D

33. [Maximum mark: 1]

18M.1.HL.TZ1.30

Four identical, positive, point charges of magnitude Q are placed at the vertices of a square of side $2d$. What is the electric potential produced at the centre of the square by the four charges?



- A. 0
- B. $\frac{4kQ}{d}$
- C. $\frac{\sqrt{2}kQ}{d}$
- D. $\frac{2\sqrt{2}kQ}{d}$

[1]

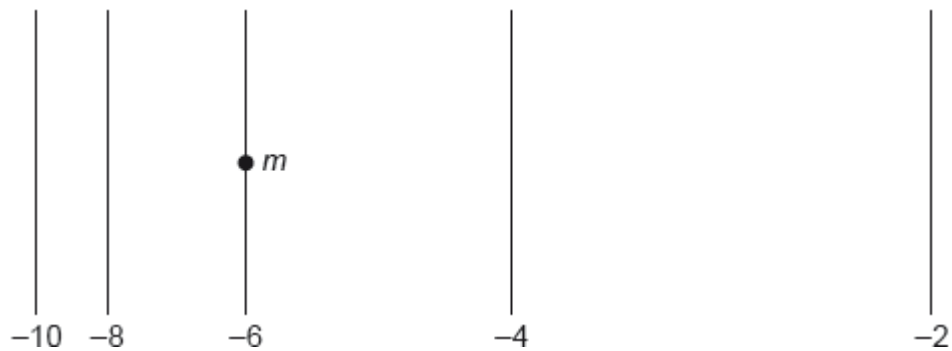
Markscheme

D

34. [Maximum mark: 1]

18M.1.HL.TZ1.31

The diagram shows 5 gravitational equipotential lines. The gravitational potential on each line is indicated. A point mass m is placed on the middle line and is then released. Values given in MJ kg^{-1} .



Which is correct about the direction of motion and the acceleration of the point mass?

	Direction	Acceleration
A.	to the right	decreasing
B.	to the right	increasing
C.	to the left	decreasing
D.	to the left	increasing

[1]

Markscheme

D

35. [Maximum mark: 1]

18M.1.HL.TZ1.32

An electron of mass m_e orbits an alpha particle of mass m_α in a circular orbit of radius r . Which expression gives the speed of the electron?

A. $\sqrt{\frac{2ke^2}{m_e r}}$

B. $\sqrt{\frac{2ke^2}{m_\alpha r}}$

C. $\sqrt{\frac{4ke^2}{m_e r}}$

D. $\sqrt{\frac{4ke^2}{m_\alpha r}}$

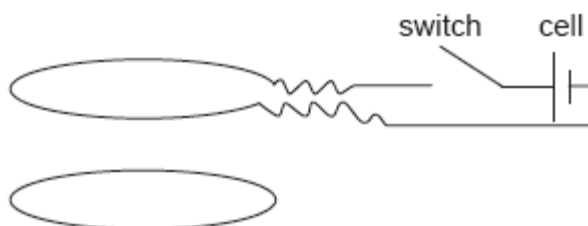
[1]

Markscheme
A

36. [Maximum mark: 1]

18M.1.HL.TZ1.33

Two identical circular coils are placed one below the other so that their planes are both horizontal. The top coil is connected to a cell and a switch.



The switch is closed and then opened. What is the force between the coils when the switch is closing and when the switch is opening?

	Switch is closing	Switch is opening
A.	attractive	attractive
B.	attractive	repulsive
C.	repulsive	attractive
D.	repulsive	repulsive

[1]

Markscheme

C

37. [Maximum mark: 1]

18M.1.HL.TZ2.28

A moon of mass M orbits a planet of mass $100M$. The radius of the planet is R and the distance between the centres of the planet and moon is $22R$.



What is the distance from the centre of the planet at which the total gravitational potential has a maximum value?

- A. $2R$
- B. $11R$
- C. $20R$
- D. $2R$ and $20R$

[1]

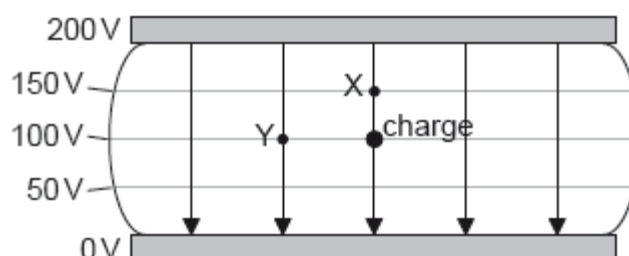
Markscheme

C

38. [Maximum mark: 1]

18M.1.HL.TZ2.29

The diagram shows the electric field and the electric equipotential surfaces between two charged parallel plates. The potential difference between the plates is 200 V.



What is the work done, in nJ, by the electric field in moving a negative charge of magnitude 1 nC from the position shown to X and to Y?

	To X	To Y
A.	50	0
B.	-50	0
C.	50	100
D.	-50	-100

[1]

Markscheme

A

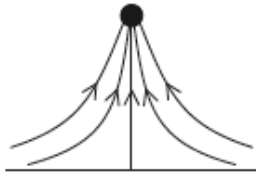
39. [Maximum mark: 1]

18M.1.HL.TZ2.30

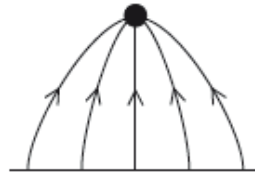
A positive point charge is placed above a metal plate at zero electric potential. Which diagram shows the pattern of electric field lines

between the charge and the plate?

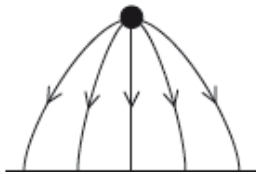
A.



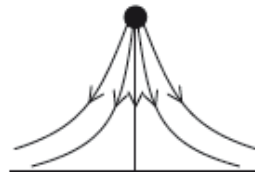
B.



C.



D.



[1]

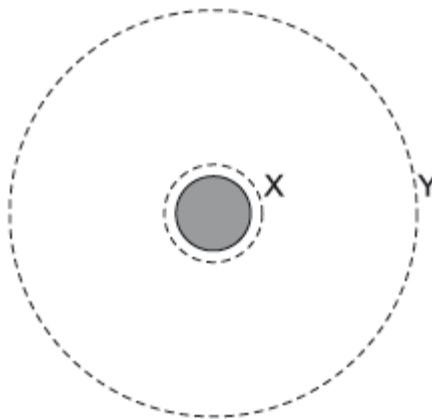
Markscheme

C

40. [Maximum mark: 1]

18M.1.HL.TZ2.31

A satellite orbiting a planet moves from orbit X to orbit Y.



What is the change in the kinetic energy and the change in the gravitational potential energy as a result?

	Kinetic energy	Gravitational potential energy
A.	increases	increases
B.	increases	decreases
C.	decreases	increases
D.	decreases	decreases

[1]

Markscheme

C

41. [Maximum mark: 1]

18M.1.HL.TZ2.32

The mass of the Earth is M_E and the mass of the Moon is M_M . Their respective radii are R_E and R_M .

Which is the ratio $\frac{\text{escape speed from the Earth}}{\text{escape speed from the Moon}}$?

A. $\sqrt{\frac{M_M R_M}{M_E R_E}}$

B. $\sqrt{\frac{M_E R_E}{M_M R_M}}$

C. $\sqrt{\frac{M_E R_M}{M_M R_E}}$

D. $\sqrt{\frac{M_M R_E}{M_E R_M}}$

[1]

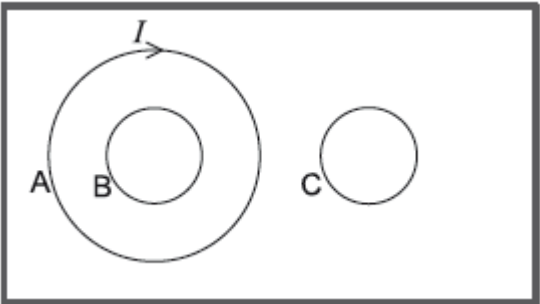
Markscheme

C

42. [Maximum mark: 1]

18M.1.HL.TZ2.33

The current flowing in loop A in a clockwise direction is increasing so as to induce a current both in loops B and C. All three loops are on the same plane.



What is the direction of the induced currents in loop B and loop C?

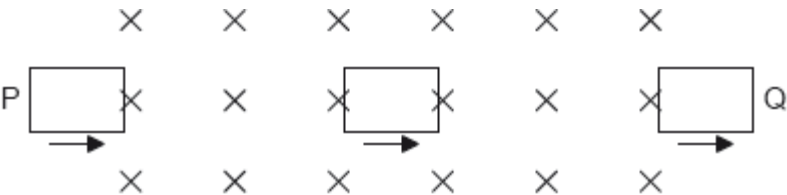
	Loop B	Loop C
A.	clockwise	clockwise
B.	clockwise	anti-clockwise
C.	anti-clockwise	clockwise
D.	anti-clockwise	anti-clockwise

[1]

Markscheme
C

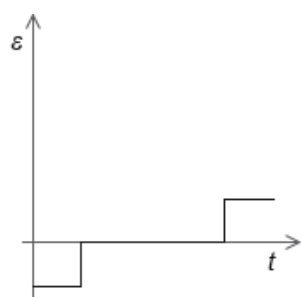
43. [Maximum mark: 1] 18M.1.HL.TZ2.34

A rectangular flat coil moves at constant speed through a uniform magnetic field. The direction of the field is into the plane of the paper.

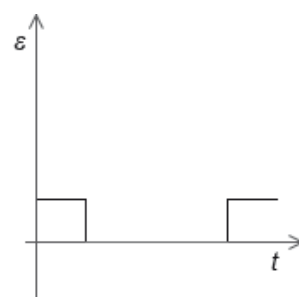


Which graph shows the variation with time t , of the induced emf ε in the coil as it moves from P to Q?

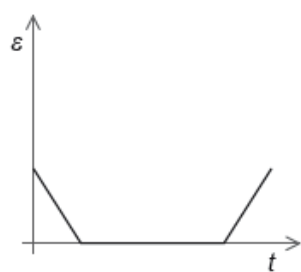
A.



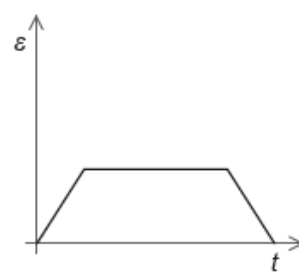
B.



C.



D.



[1]

Markscheme

A